

KARTHIK V

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EDUCATION

Chennai Institute of Technology

Bachelor of Technology in Computer Science and Business Systems [CGPA - 8.3]

Chennai, India

September 2023 – May 2027

EXPERIENCE

Research Intern

Chennai Institute of Technology

Nov 2024 – Dec 2024

Chennai, India

- BLOCKCHAIN MESSAGING APPLICATION USING SOLANA WITH ML-KEM(Research)(Under Review)
- AMQP IN IOT FOCUSING ON ENERGY CONSUMPTION AND EFFICIENCY(Review)(Under Review)

PROJECTS

Brain_Tumor_Detection_and_Pill_identification_using_CNN_BioBert | Streamlit, RAG, OCR, Flask

- This project leverages (CNNs) for **brain tumor** detection from medical imaging and **BioBERT** for **pill** identification and classification based on textual and image-based inputs.
- **AI Tumor Detection**: Detects brain tumors from medical images and provides detailed AI-generated diagnostic reports.
- **AI Pill Detector**: Identifies pills based on images and provides detailed information, including name, usage, dosage, and purpose.

Data-Stats-and-Visualization-Generation-Using-NLP | React JS, Python, Gemini API, Node JS

- This project allows users to upload a **CSV** file, query insights from the dataset, and get visuals recommendations based on those queries.
- It uses NLP and generative AI to extract meaningful insights, predict the need for visual representation (charts/graphs), and generate Python code that visualizes the data using libraries like matplotlib.

GpsSos| Kotlin, NginX, Python (Flask), MongoDB, OSM-MapView

- Continuously logs GPS location locally and syncs to server via Flask backend when online, using MongoDB for storage.
- Planned to log emergency data like device states, enhancing real-time situational awareness in critical events.

CERTIFICATIONS

- **Remote Sensing** | ISRO
- **AWS Cloud Practitioner** | AWS
- **Database Management System** | NPTEL
- **Cloud Computing** | NPTEL
- **Google Cybersecurity Specialist** | Google Coursera
- **Hacking and patching** | Colorado university Coursera
- **Linux command and shell scripting** | IBM Coursera

PLATFORMS

- **CodeForce**: 381 Peak rating
- **Codechef**: 1132 Peak rating
- **Leetcode**: 1730 Peak rating, 509 problems

TECHNICAL SKILLS

Languages: Python, Kotlin, C++, SQL, Bash, HTML/CSS, Java, JavaScript, Kotlin.

Tools: Burp Suite, WireShark, Nmap, Kali, Blender, AfterEffects

Technologies: Node.js, Flask, Django, RabbitMQ&AMQP, MongoDB, React, ReactNative

Developer Tools: Linux, Git, AWS, Google Cloud & Colab, VS Code, Kubernetes, Docker, Vim, Github

Domain Knowledge: Security Platform/Tool Development, Threat Research, Open Source Intelligence, Web Monitoring, Pen Test Automation

ACHIEVEMENTS

Appian AI Challenge – IIT Madras | 5th Runner-Up: Pitched our Brain_Tumor_Detection_and_Pill_identification_using_CNN_BioBert idea.

VIT Web3 Edition - VIT Chennai: Presented our all in one Healthcare solution Healix.

NAF Uzhavu Hackathon – IIT Madras | Finalist: Developed IoT-based smart agriculture solutions for precision farming.

Hack With Amazon– Amazon | Progressed to 2/3 level by solving aptitude quiz.

VOLUNTEERING

Japanese Consultant Event | Contributed making Poster Designing for Event

Zorphix Web Development Team | Design UI, Backend and Animation for symposium website.